

ANSWER SCHEME

SECTION A		
1.	(a)	$\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 2x} = \lim_{x \rightarrow 2} \frac{(x-2)(x^2 + 2x + 4)}{x(x-2)} = \lim_{x \rightarrow 2} \frac{x^2 + 2x + 4}{x} = \frac{2^2 + 2(2) + 4}{2} = 6$
	(b)	$\lim_{x \rightarrow \infty} \sqrt{\frac{5x+7}{6x-5}} = \lim_{x \rightarrow \infty} \sqrt{\frac{\frac{5x}{x} + \frac{7}{x}}{\frac{6x}{x} - \frac{5}{x}}} = \lim_{x \rightarrow \infty} \sqrt{\frac{5 + \frac{7}{x}}{6 - \frac{5}{x}}} = \sqrt{\frac{5+0}{6-0}} = \sqrt{\frac{5}{6}}$
	(c)	$\lim_{x \rightarrow 3^-} \frac{ 4x-12 }{x-3} = \lim_{x \rightarrow 3^-} \frac{-4(x-3)}{x-3} = \lim_{x \rightarrow 3^-} -4 = -4$
2.	(a)	$g(x) = (2-x)^3(x-1)$ $g'(x) = uv' + vu'$ $g'(x) = (2-x)^3(1) + (x-1)3(2-x)^2(-1)$ $g'(x) = (2-x)^2[(2-x) - 3(x-1)]$ $g'(x) = (2-x)^2(5-4x)$
	(b)	$y = e^{-2x} \sin 3x$ $\frac{dy}{dx} = -2e^{-2x} \sin 3x + 3e^{-2x} \cos 3x$ $= e^{-2x}(-2 \sin 3x + 3 \cos 3x)$ $\frac{d^2y}{dx^2} = -2e^{-2x}(-2 \sin 3x + 3 \cos 3x) + e^{-2x}(-6 \cos 3x - 9 \sin 3x)$ $= 4e^{-2x} \sin 3x - 6e^{-2x} \cos 3x - 6e^{-2x} \cos 3x - 9e^{-2x} \sin 3x$ $= -5e^{-2x} \sin 3x - 12e^{-2x} \cos 3x$ $\frac{d^2y}{dx^2} + 4 \frac{dy}{dx} + 13y$ $= -5e^{-2x} \sin 3x - 12e^{-2x} \cos 3x + 4e^{-2x}(-2 \sin 3x + 3 \cos 3x) + 13e^{-2x} \sin 3x$ $= 0(\text{shown})$
3.		$\frac{dA}{dt} = -2 \text{ cm}^2 / \text{min}$ $\frac{dV}{dt} = ? \text{ when } r = 5 \text{ cm}$ $A = 4\pi r^2 \Rightarrow \frac{dA}{dr} = 8\pi r$

		$V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \cdot \frac{dr}{dt}$ $\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$ $-2 = 8\pi(5) \times \frac{dr}{dt}$ $\frac{dr}{dt} = \frac{-2}{40\pi} = -\frac{1}{20\pi} \text{ cm/min}$ $\therefore \frac{dV}{dt} = 4\pi(5)^2 \cdot \frac{-1}{20\pi} = -5 \text{ cm}^3 / \text{min}$
SECTION B		
1.	(a)	$\bar{z} - \frac{1}{\bar{z}} = (2-i) - \frac{1}{(2-i)}$ $= (2-i) - \frac{1}{(2-i)} \cdot \frac{(2+i)}{(2+i)}$ $= (2-i) - \frac{(2+i)}{(2^2+1)}$ $= \frac{5(2-i) - 2-i}{5}$ $= \frac{8}{5} - \frac{6}{5}i$
	(b)	$\left \bar{z} - \frac{1}{\bar{z}} \right = \sqrt{\left(\frac{8}{5}\right)^2 + \left(-\frac{6}{5}\right)^2}$ $= \sqrt{\frac{64}{25} + \frac{36}{25}}$ $= \sqrt{4}$ $= 2$
2.	(a)	$3\log_9 x = (\log_3 x)^2$ $\frac{3\log_3 x}{\log_3 9} = (\log_3 x)^2$ $\frac{3\log_3 x}{\log_3 3^2} = (\log_3 x)^2$ $\frac{3\log_3 x}{2} = (\log_3 x)^2$ $\text{Let } u = \log_3 x \therefore \frac{3u}{2} = u^2$ $2u^2 - 3u = 0$

		$u(2u - 3) = 0$ $u = 0 \text{ or } u = \frac{3}{2}$ Therefore $\log_3 x = 0 \Rightarrow x = 3^0 = 1$ $\log_3 x = \frac{3}{2} \Rightarrow x = 3^{\frac{3}{2}} = \sqrt{27}$												
	(b)	Given $\frac{5}{x+3} - 1 \leq 0$ $\frac{2-x}{x+3} \leq 0$ <table border="1"><tr><td>$2-x$</td><td>+</td><td>+</td><td>-</td></tr><tr><td>$x+3$</td><td>-</td><td>+</td><td>+</td></tr><tr><td>$\frac{2-x}{x+3}$</td><td>-</td><td>+</td><td>-</td></tr></table> $\therefore \{x : x < -3 \cup x \geq 2\}$	$2-x$	+	+	-	$x+3$	-	+	+	$\frac{2-x}{x+3}$	-	+	-
$2-x$	+	+	-											
$x+3$	-	+	+											
$\frac{2-x}{x+3}$	-	+	-											
3.	(a)	$g(x_1) = g(x_2)$ $\frac{1}{2x_1 - 5} = \frac{1}{2x_2 - 5}$ $2x_2 - 5 = 2x_1 - 5$ $2x_2 = 2x_1$ $x_2 = x_1$ $\therefore g(x)$ is one to one function $g \circ g^{-1}(x) = x$ $g(g^{-1}(x)) = x$ $\frac{1}{2g^{-1}(x) - 5} = x$ $1 = x(2g^{-1}(x) - 5)$ $1 = 2xg^{-1}(x) - 5x$ $1 + 5x = 2xg^{-1}(x)$ $g^{-1}(x) = \frac{1+5x}{2x} = \frac{1}{2x} + \frac{5}{2}$												

	(b)	$g \circ g^{-1}(x)$ $= g(g^{-1}(x))$ $= g\left(\frac{1+5x}{2x}\right)$ $= \frac{1}{2\left(\frac{1+5x}{2x}\right) - 5}$ $= \frac{1}{\frac{1+5x}{x} - 5}$ $= \frac{1}{\frac{1}{x}}$ $= x \text{ (show)}$
4.	(a)	To find $f^{-1}(x)$: $f \circ f^{-1}(x) = x$ $(f^{-1}(x) - 2)^2 + 1 = x$ $(f^{-1}(x) - 2)^2 = x - 1$ $f^{-1}(x) - 2 = \pm\sqrt{x-1}$ $f^{-1}(x) = \pm\sqrt{x-1} + 2$ $x > 2, \therefore f^{-1}(x) = \sqrt{x-1} + 2$ To find $g^{-1}(x)$: $g \circ g^{-1}(x) = x$ $\ln(g^{-1}(x) + 1) = x$ $g^{-1}(x) + 1 = e^x$ $g^{-1}(x) = e^x - 1$ $D_{f^{-1}} = \{x > 1\}, R_{f^{-1}} = \{y > 2\}$ $D_{g^{-1}} = \{x > 0\}, R_{g^{-1}} = \{y > 0\}$
	(b)	$g(f(x)) = g((x-2)^2 + 1) = \ln(((x-2)^2 + 1) + 1) = \ln((x-2)^2 + 2)$ $g(f(2)) = \ln((2-2)^2 + 2) = \ln 2$

5.	$f(x)$ is continuous at $x = a$ if all the following three conditions are fulfilled : (i) $f(a)$ is defined (ii) $\lim_{x \rightarrow a} f(x)$ exists, $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x)$ (iii) $f(a) = \lim_{x \rightarrow a} f(x)$
(a)	$f(x)$ is continuous at $x = 0$ $f(0) = \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x)$ $2 = \lim_{x \rightarrow 0^-} (-2 \cos x + n) = \lim_{x \rightarrow 0^+} (mx^2 + 2)$ $2 = -2 + n = 2$ $n = 4$ $f(x)$ is continuous at $x = 2$ $f(2) = \lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x)$ $-2 + m = \lim_{x \rightarrow 2^-} (mx^2 + 2) = \lim_{x \rightarrow 2^+} (-x + m)$ $-2 + m = 4m + 2 = -2 + m$ $3m = -4$ $m = \frac{-4}{3}$
(b)	$f(x) = \begin{cases} 4 - 2 \cos x, & x < 0 \\ 2 - 2x^2, & 0 \leq x < 2 \\ -2 - x, & x \geq 2 \end{cases}$ $f'(2) = \lim_{x \rightarrow 2^+} \frac{f(x) - f(2)}{x - 2}$ $= \lim_{x \rightarrow 2^+} \frac{(-2 - x) - (-4)}{x - 2}$ $= \lim_{x \rightarrow 2^+} \frac{-x + 2}{x - 2}$ $= \lim_{x \rightarrow 2^+} \frac{-(x - 2)}{x - 2}$ $= \lim_{x \rightarrow 2^+} -1$ $= -1$ $f'(2) = \lim_{x \rightarrow 2^-} \frac{f(x) - f(2)}{x - 2}$ $= \lim_{x \rightarrow 2^-} \frac{(2 - 2x^2) - (-4)}{x - 2}$ $= \lim_{x \rightarrow 2^-} \frac{-2x^2 + 6}{x - 2}$ $= \lim_{x \rightarrow 2^-} \frac{-2(x^2 - 3)}{x - 2}$ $= \text{undefined}$ since $\lim_{x \rightarrow 2^+} f(x) \neq \lim_{x \rightarrow 2^-} f(x)$, so $f(x)$ is not differentiable at $x = 2$

6.	(a)	$x = 3t - \frac{1}{t} \qquad y = t + \frac{3}{t}$ $\frac{dx}{dt} = 3 + \frac{1}{t^2} = \frac{3t^2 + 1}{t^2} \qquad \frac{dy}{dt} = 1 - \frac{3}{t^2} = \frac{t^2 - 3}{t^2}$ $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} = \frac{t^2 - 3}{t^2} \times \frac{t^2}{3t^2 + 1} = \frac{t^2 - 3}{3t^2 + 1}. \quad (\text{shown})$ $\frac{d}{dt} \left(\frac{dy}{dx} \right) = \frac{(3t^2 + 1)(2t) - (t^2 - 3)(6t)}{(3t^2 + 1)^2} = \frac{20t}{(3t^2 + 1)^2}$ $\frac{d^2 y}{dx^2} = \frac{d}{dt} \left(\frac{dy}{dx} \right) \times \frac{dt}{dx} = \frac{20t}{(3t^2 + 1)^2} \times \frac{t^2}{3t^2 + 1} = \frac{20t^3}{(3t^2 + 1)^3}$
	(b)	$x^2 - 3y^2 = ae^{y-2x} + by - 6 \quad \dots\dots\dots(1)$ Substitute $x = 1, y = 2$ into (1) $1^2 - 3(2)^2 = ae^{2-2(1)} + b(2) - 6$ $a + 2b = -5 \quad \dots\dots\dots(2)$ Differentiate with respect to x implicitly equation (1) $2x - 6y \frac{dy}{dx} = ae^{y-2x} \left(\frac{dy}{dx} - 2 \right) + b \frac{dy}{dx} \quad \dots\dots\dots(3)$ Substitute $x = 1, y = 2$ and $\frac{dy}{dx} = 1$ into (3) $2 - 6(2)(1) = a(1)(1 - 2) + b(1)$ $-10 = -a + b \quad \dots\dots\dots(4)$ (2) + (4) $3b = -15$ $b = -5$ Substitute $b = -5$ into (4) $a = 5$

7.	(a)	$f(x) = \frac{\ln(x-1)}{x-1}$ $f'(x) = \frac{\frac{1}{x-1}(x-1) - \ln(x-1)}{(x-1)^2}$ $= \frac{1 - \ln(x-1)}{(x-1)^2}$ $f''(x) = \frac{-\frac{1}{x-1}(x-1)^2 - (1 - \ln(x-1))(2)(x-1)}{(x-1)^4}$ $= \frac{-(x-1) - 2(x-1)(1 - \ln(x-1))}{(x-1)^4}$ $= \frac{(x-1)(-1 - 2 + 2\ln(x-1))}{(x-1)^4}$ $= \frac{2\ln(x-1) - 3}{(x-1)^3}$ Stationary point when $f'(x) = 0$ $1 - \ln(x-1) = 0$ $x - 1 = e$ $x = e + 1$ $f''(e+1) = \frac{2\ln(e+1-1) - 3}{(e+1-1)^3} = \frac{2\ln(e) - 3}{e^3} = \frac{-1}{e^3} < 0$ There is only one maximum point of $f(x)$
	(b)	When $x = e + 1$, $f(e+1) = \frac{\ln(e+1-1)}{e+1-1} = \frac{\ln(e)}{e} = \frac{1}{e}$ The exact coordinate is $\left(e+1, \frac{1}{e}\right)$